

Evaluating Hybrid Post-Quantum TLS Resilience on Constrained IoT Devices Under Bursty Network Conditions.

No Author Given

No Institute Given

Abstract. Transport Layer Security (TLS), particularly hybrid PQC-TLS (PQC), is emerging as the key migration strategy to help secure Internet of Things (IoT) communication channels in the face of future quantum adversaries. The major increase in the size of the key exchange and digital signature payloads of post-quantum algorithms, however, is very concerning for resource-limited embedded devices on lossy wireless channels. Existing works on PQC-TLS have typically been based on a uniform and independent model of packet loss, which is not representative of the bursty and time-correlated failures of common IoT connections like LoRaWAN, Cellular-IoT, and overloaded Wi-Fi. This is an important difference because fragmented handshakes are inevitable when using large PQC public keys and certificates, and these multi-packet handshakes are extremely susceptible to loss of subsequent packets. This paper compares an implementation of hybrid PQC-TLS 1.3 (using X25519 for key exchange and ML-KEM-768 for the key exchange, and using ML-DSA-44 for authentication), with an implementation of classical TLS 1.3 on an ESP32 microcontroller in an empirical evaluation. A two-state Gilbert-Elliott burst-loss model of network degradation is emulated, calibrated according to real-world wireless packet trace captures, and with controlled latency and jitter injection. For both the TCP and DTLS 1.3 transport layers, we measure handshake completion times, failure and timeout rates, retransmission overhead, total byte counts and per-session energy consumption with calibrated current-sensing hardware. In addition, we test TLS 1.3 Session Resumption as a practical mitigation and quantify how much of the PQC performance penalty it is able to recover in the case of a severe burst loss. Initial experiments show that bursty loss significantly reduces the completion rate of hybrid PQC-TLS handshake compared to uniform loss at a similar mean loss probability, and that the tail-latency of the hybrid handshake is severely affected by IP fragmentation and transport-layer Head-of-Line (HoL) blocking in the case of TLS over TCP. Beyond that, our assessment shows that TLS 1.3 Session Resumption is able to meaningfully ameliorate this latency and energy cost, since it avoids re-transmission of large PQC public keys and certificates for future connections. These results provide quantitative deployment thresholds across a range of loss models, transport layers, and session-resumption states. It also provides concrete protocol guidance for securing unreliable wireless IoT infrastructure.

Keywords: quantum Cryptography, Hybrid TLS 1.3, IoT Security, Gilbert-Elliott Model, Packet Loss, ESP32, Energy Consumption.